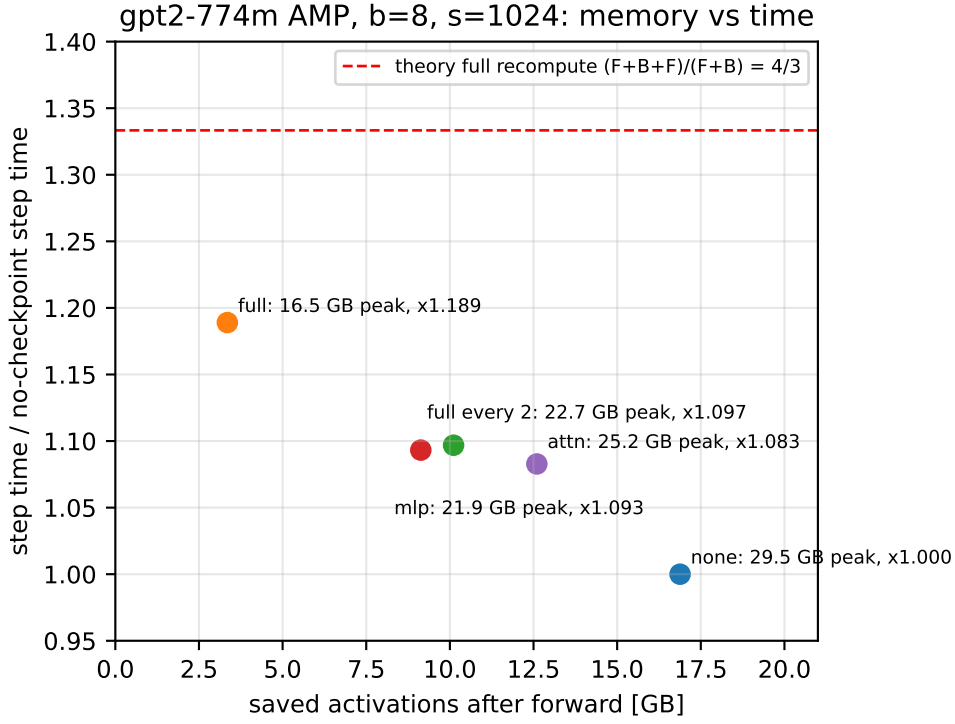
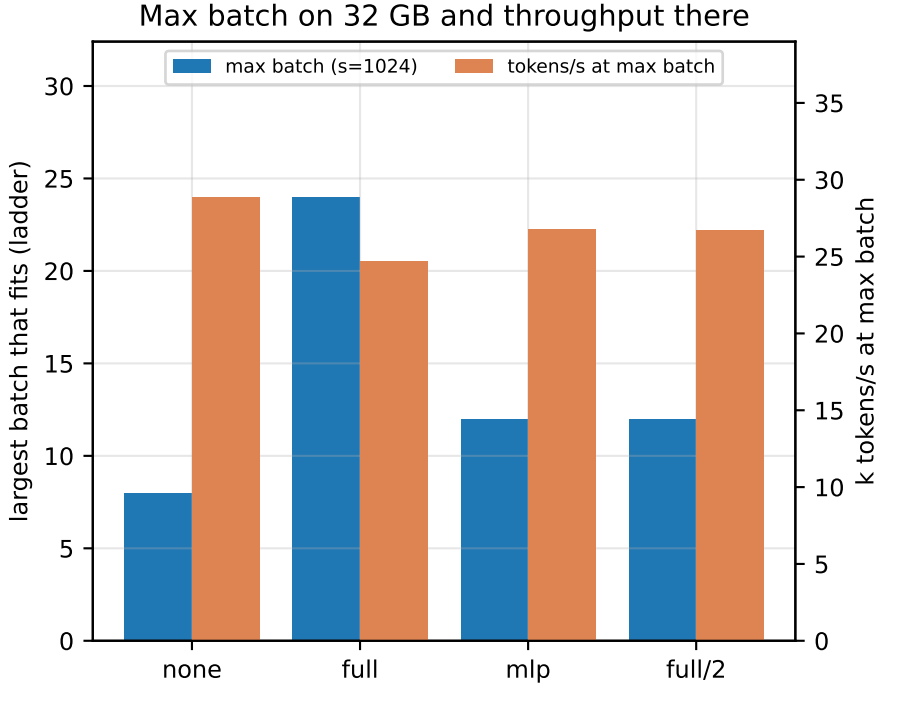

gpt2-774m AMP, b=8, s=1024: memory vs time
theory full recompute (F+B+F)/(F+B) = 4/3
full: 16.5 GB peak, x1.189
full every 2: 22.7 GB peak, x1.097
attn: 25.2 GB peak, x1.083
mlp: 21.9 GB peak, x1.093
none: 29.5 GB peak, x1.000
step time / no-checkpoint step time
saved activations after forward [GB]
1.40
1.35
1.30
1.25
1.20
1.15
1.10
1.05
1.00
0.95
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Max batch on 32 GB and throughput there
max batch (s=1024)
tokens/s at max batch
largest batch that fits (ladder)
k tokens/s at max batch
30
25
20
15
10
5
0
35
30
25
20
15
10
5
0
none
full
mlp
full/2